MoonFall: Lunar Outpost — Mission Architecture & Scientific Citations

Official NASA, NOAA & Peer-Reviewed Scientific References by Mission Module

Mission 1.1: Power Station

Subsystem Scope: Photovoltaic generation, polar siting at Shackleton Crater, battery storage, and fission surface power.

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- [4] Vertical Solar Array Technology (VSAT) — NASA TechPort — https://techport.nasa.gov/projects/116305
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- [8] Lunar Surface Technology Overview — NASA — https://www.nasa.gov/lunar-surface-technology/
- [9] To the Moon and Back: Lunar Surface Environment & Day/Night Cycles — NASA JSC — https://ares.jsc.nasa.gov/interaction/lmdp/documents/TotheMoon_back.pdf
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- [13] Fission Surface Power Project Overview — NASA ESDMD — https://www.nasa.gov/exploration-systems-development-mission-directorate/fission-surface-power/
- [14] Electric Power on the Moon — NASA & JAXA Technical Paper — https://ntrs.nasa.gov/api/citations/20250000763/downloads/Csank_Pr_LunarSurfacePowerJAXA20250000763.pdf
- [15] NASA Power Research & Architecture — https://www.nasa.gov/power/
- [16] Powering the Moon Using Vertical Solar Arrays (NASA Video) — https://www.youtube.com/watch?v=TbVIMSVuEFU
- [17] Power and Energy for the Lunar Surface — NASA NTRS — https://ntrs.nasa.gov/api/citations/20220005893/downloads/Power%20and%20Energy%20for%20the%20Lunar%20Surface.pdf
- [18] Ames Technical Standard (NASA STD 8070.1) — https://www.nasa.gov/wp-content/uploads/2017/03/std8070.1.pdf
- [19] Guidelines on Lithium-ion Battery Use in Space Applications — NASA NTRS — https://ntrs.nasa.gov/api/citations/20090023862/downloads/20090023862.pdf

Mission 2.1: Daily Meal & Food Storage

Subsystem Scope: Lunar food storage, multilayer pouch packaging, preservation atmospheres, and crew nutrition.

- [1] 7.0 Habitability Functions — NASA Human Integration Standards — https://www.nasa.gov/reference/7-0-habit-ability-functions-vol-2/
- [2] Space Food Packaging Facts Brochure — NASA — https://www.nasa.gov/wp-content/uploads/2024/06/space-food-packagingbrochure.pdf?emrc=6a0ea429179ff
- [3] Food and Nutrition Technical Brief — NASA OCHMO — https://www.nasa.gov/wp-content/uploads/2023/03/food-and-nutrition-technical-brief-ochmo.pdf
- [4] Space Food Fact Sheet — NASA — https://www.nasa.gov/wp-content/uploads/2015/05/167750main_fs_spacefood508c.pdf
- [5] Food in Space — NASA Office of the Chief Health & Medical Officer — https://www.nasa.gov/ochmo/food-in-space/
- [6] Crew Systems Overview — NASA — https://www.nasa.gov/reference/crew-systems/
- [7] NASA Space Food Formulations: Rehydratable Foods — https://www.nasa.gov/wp-content/uploads/2024/06/nasa-space-food-formulations-rehydratable.pdf?emrc=6ae09bc88e6f0

Mission 2.2: Hydroponics & Lunar Plant Growth

Subsystem Scope: Lunar regolith simulant testing, dust hazards, hydroponics cultivation, and plant gene expression.

- [1] Scientists Grow Plants in Lunar Soil — NASA — https://www.nasa.gov/humans-in-space/scientists-grow-plants-in-lunar-soil/
- [2] Moon Farming — NASA Houston We Have a Podcast — https://www.nasa.gov/podcasts/houston-we-have-a-podcast/moon-farming/
- [3] What Is Lunar Regolith? The Moon's Surface Explained — NASA Science — https://science.nasa.gov/biological-physical/what-is-lunar-regolith/
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- [6] Lunar Regolith Simulant User's Guide (Rev A) — NASA NTRS — https://ntrs.nasa.gov/api/citations/20240011783/downloads/Lunar_Regolith_Simulant_Users_Guide_Rev_A_28OCT.pdf
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- [8] Growing Plants in Lunar Soil — NASA Science — https://science.nasa.gov/resource/growing-plants-in-lunar-soil/
- [9] NASA Open Science Data Repository (OSD-476 Botanical Study) — https://osdr.nasa.gov/bio/repo/data/studies/OSD-476
- [10] Scientists Grew Plants in Moon Soil — What We Learned (NASA Video) — https://www.youtube.com/watch?v=W5OzSeTHouU

- [11] How to Grow Food on the Moon — NASA Gravity Assist Podcast — https://www.nasa.gov/podcasts/gravity-assist/gravity-assist-how-to-grow-food-on-the-moon/

Mission 5.1: Space Radiation Shelter

Subsystem Scope: Solar Particle Events (SPEs), Galactic Cosmic Rays (GCRs), in-situ regolith mass shielding, and shelter engineering.

- [1] Design for Ionizing Radiation Protection (OCHMO-TB-020 Rev G) — NASA — https://www.nasa.gov/wp-content/uploads/2023/12/ochmo-tb-020-radiation-protection.pdf
- [2] Where on the Moon: Radiation Environment — Apollo Lunar Surface Journal — https://www.nasa.gov/wp-content/uploads/static/history/alsj/WOTM/WOTM-Radiation.html
- [3] Effects of Space Radiation — Small Steps, Giant Leaps (Ep90) — NASA — https://www.nasa.gov/podcasts/small-steps-giant-leaps/small-steps-giant-leaps-episode-90-effects-of-space-radiation/
- [4] Glenn Symposium Radiation Protection Panel Charts — NASA NTRS — https://ntrs.nasa.gov/api/citations/20210018122/downloads/Glenn%20Symposium%20Panel%20charts_2021%20-%20Jonathan%20Ransom.pdf
- [5] Protecting Astronauts from Space Radiation on the Lunar Surface — NASA NTRS — https://ntrs.nasa.gov/api/citations/20210009760/downloads/Clowdsley_Lunar_Radiation_Talk_v2.pdf
- [6] Protecting Crew and Surface Systems with Lunar Safe Haven — ASCEND 2021 — https://ntrs.nasa.gov/api/citations/20210022066/downloads/Grande_Lunar%20Safe%20Haven_ASCEND%202021_Full_v2b.docx.pdf
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- [9] Lunar In-Situ Resource Utilization (Kleinhenz, ASCE 2021) — https://ntrs.nasa.gov/api/citations/20210012921/downloads/ASCE_ES_2021_shortCourse_ISRU_Kleinhenz_v2.pdf?attachment=true

Mission 3: ISRU Facility

Subsystem Scope: Lunar ice cold traps, TRIDENT subsurface drilling, MSolo mass spectrometry, water extraction, and OGA electrolysis.

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- [2] Water on the Moon — NASA Goddard (Video) — https://svs.gsfc.nasa.gov/11756
- [3] The Moon's South Pole & Permanent Shadows — NASA Goddard SVS — https://svs.gsfc.nasa.gov/gallery/moonpole
- [4] Is There Water on the Moon? We Asked a NASA Scientist (Ep27) — https://www.nasa.gov/general/is-there-water-on-the-moon-we-asked-a-nasa-scientist-episode-27
- [5] Apollo to Artemis: Drilling on the Moon — NASA Kennedy — https://www.nasa.gov/centers-and-facilities/kennedy/apollo-to-artemis-drilling-on-the-moon/
- [6] PRIME-1 Science & Technology (TRIDENT Drill and MSolo) — NASA Science — https://science.nasa.gov/lunar-science/clps-deliveries/prime-1-im/

- [7] NASA's Lunar Drill Technology Passes Tests on the Moon — https://www.nasa.gov/missions/artemis/clps/nasas-lunar-drill-technology-passes-tests-on-the-moon/
- [8] PRIME-1 Lunar Drill Video — NASA — https://www.nasa.gov/video-detail/prime-1-2025-updated-video/
- [9] Overview: In-Situ Resource Utilization (ISRU) — NASA — https://www.nasa.gov/overview-in-situ-resource-utilization/
- [10] NASA Glenn Excavation Laboratory — https://www.nasa.gov/?p=38643
- [11] Integration of ISRU Into Lunar/Mars Exploration (Field Analogs & Reactor) — https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20100021362.pdf
- [12] Low Pressure Frost Formation for Lunar Polar Water Capture — NASA NTRS — https://ntrs.nasa.gov/citations/20230010063
- [13] Microwave Extraction of Water from Lunar Regolith Simulant — NASA NTRS — https://ntrs.nasa.gov/api/citations/20070018802/downloads/20070018802.pdf
- [14] How Will We Extract Water on the Moon? We Asked a NASA Technologist (Ep47) — https://www.nasa.gov/general/how-will-we-extract-water-on-the-moon-we-asked-a-nasa-technologist-episode-47
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- [16] Electrochemical Solutions for Advanced Life Support (ISS OGA Electrolysis) — NASA NTRS — https://ntrs.nasa.gov/citations/20200002346
- [17] ISS Environmental Control and Life Support System Fact Sheet — NASA Marshall — https://www.nasa.gov/sites/default/files/atoms/files/g-281237_eclss_0.pdf

Final Mission: MoonFall Last Light (The 18-Hour Lunar Crisis)

Subsystem Scope: Eight-stage emergency survival sequence, life support triage, power distribution, and speed-of-light communication latency.

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- [11] Food and Nutrition Technical Brief — NASA Office of the Chief Health & Medical Officer (OCHMO) — https://www.nasa.gov/wp-content/uploads/2023/03/food-and-nutrition-technical-brief-ochmo.pdf
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- [13] Lunar Laser Ranging: A Continuing Legacy of the Apollo Program (Earth-Moon Distance & Light Latency) — https://ntrs.nasa.gov/citations/20210004132
- [14] NASA Spaceflight Human-System Standard Volume 2 Rev E: Human Factors, Habitability, and Environmental Health — https://ntrs.nasa.gov/citations/20250004555
- [15] Overview: In-Situ Resource Utilization (ISRU) — NASA Exploration Systems — https://www.nasa.gov/overview-in-situ-resource-utilization/